NEW YORK COLLEGE
THE INTERNATIONAL COLLEGE OF GREECE

University of Greater Manchester

BSc(HONS)COMPUTING

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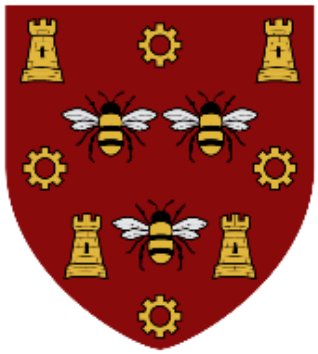
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System analysis and design

2024/2025

Semester 1

Cohort 2



LEVEL HE4

BSc(HONS)COMPUTING

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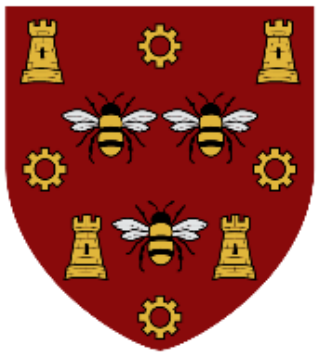
System analysis and design

MODULE GUIDE

2024/2025

Semester 1

Cohort 2



LEVEL HE4

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1. Assessment Overview

Module Tutor	Dr.Georgios Papamichail
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Weblink to Moodle Class	https://elearning.nyc.gr/Education/ViewLessonStructure.aspx?lang=el-GR&lessonID=5206&classID=1465
Weblink to Module Specification	https://elearning.nyc.gr/Education/ViewLessonStructure.aspx?lang=el-GR&lessonID=5206&classID=1465
Deadline	23/12/2025

2. Assessment Outline

You will produce a portfolio of tasks related to networking, subnetting and proposing a networking solution.

3. Introduction

You are working as a network consultant for a company that manages a large educational institution offering computer science training. You need to design the organization's networking topology, VLAN design and IP Addressing scheme.

4. Assignment

Part I - IP Addressing & Subnetting – 100 points

Which is the (class) of the following IP addresses (6 points)

148.17.9.1 CLASS B

193.42.1.1 CLASS C

126.8.156.0 CLASS A

Write down the default subnet mask for the following IP addresses (6 points)

191.249.234.191 255.255.0.0

223.23.223.109 255.255.255.0

10.10.250.1 255.0.0.0

Subnetting Exercise 1 (18 points)

Network Address: **165.59.0.0/16**

Number of needed useable subnets: **1000**

Address class (2 points) B

Default subnet mask (2 points) 255.255.0.0

Number of bits borrowed (3 points) 10

Custom subnet mask (3 points) 255.255.255.192

Total number of subnets (2 points) 1024

Total number of hosts/subnet (2 points) 64

Number of usable hosts/subnet (2 points) 64-2=62

Subnetting Exercise 2 (70 points)

Network Address: 19.0.0.0/8

Number of needed useable hosts: 1000

Address class (2 points) A

Default subnet mask (2 points) 255.0.0.0

Number of bits borrowed (3 points) 14

Custom subnet mask (3 points) 255.255.252.0

Total number of subnets (2 points) 2^14 = 16384

Total number of hosts/subnet (2 points) 2^10 = 1024

Number of usable hosts/subnet (2 points) 1024-2

What is the subnet number for the 100th subnet? (13 points)

The subnet number for the 100th number is: 19.1.140.0/22

What is the 200th useable subnet host range? (13 points)

200th subnet network address: 19.3.28.0

First host: 19.3.28.1

Broadcast address:19.3.31.255

Last host: 19.3.31.254

What is the broadcast address for the 429th useable subnet? (13 points)

The broadcast address for the 429th useable subnet is 19.6.179.255

In which subnet does the address 19.0.199.172 belong, and what exactly represents? (13 points)
The address 19.0.199.172 is a usable host with the subnet 19.0.196.0/22

Part II – Network Topology – 100 points

Study the following topology and answer the following questions

1. If HOST2 sends a message to HOST6, who else will receive it and why? _(15 points)

   Only HOST 6 will receive the message because switches forward only to the correct port and routers separate broadcast domains

2. If HOST5 sends a message to HOST3, who else will receive it and why? (15 points)

   If HOST5 sends a message to HOST3 only HOST3 will receive it because the communication is unicast and routers isolate broadcast domains

3. If HOST4 sends a message to HOST10, who else will receive it and why? (15 points)

   If HOST4 sends a message to HOST 10 only HOST 10 will receive it because the switches wont flood unicast traffic and routers will forward only where needed

4. Which IP address would you put as a default gateway on HOST5 and why; (10 points)

   The default gateway for HOST5 should be 199.1.1.40 because it is the router interface in the same subnet and provides access to remote networks

5. If Host 9 sends a broadcast message who will receive it and why? (15 points)

   If HOST9 sends a broadcast message it will be received by HOST 6,HOST7,HOST 10,RTB because they are in the same subnet (199.3.3.0/24) and because broadcasts are flooded by a broadcast domain and blocked routers

6. How many Collisions Domains and how many Broadcast Domains exist in the topology? Justify your answer. (15 points)

   In the topology there are 4 broadcast domains:

   i)199.1.1.16/28

   ii)199.1.1.32/28

   iii)199.2.2.0/24

   iii)199.3.3.5/24

   For i)(host1-host3+RTA)

   Host1 -> 199.1.1.17/28

   Host 1 in binary:

   1100111.00000001.00000001.00010001

   Subnet mask-> 255.255.255.240

   Subnet mask in binary:

   11111111.11111111.11111111.11110000

   Bitwise and

   IP: 1100111.00000001.00000001.00010001

   MASK:11111111.11111111.11111111.11110000

   NET: 11000111.00000001.00000001.00010000 = 199.1.1.16/28 NETWORK ADDRESS

   For ii) (host4-host5 +RTA)

HOST4->199.1.1.33/28

HOST 4 in binary: 11000111.00000001.00000001.00100001

Bitwise AND

IP: 11000111.00000001.00000001.00100001

MASK: 11111111.11111111.11111111.11110000

NET: 11000111.00000001.00000001.00100000 = 199.1.32/28 NETWORK ADDRESS

For iii) (RTA<->RTB)

RTA -> 199.2.2.1/24

RTA in binary : 11000111.00000010.00000010.00000001

Bitwise AND

IP: 11000111.00000010.00000010.00000001

MASK: 11111111.11111111.11111111.11110000

NET:11000111.00000010.00000010.00000000 = 199.2.2.0/24 NETWORK ADDRESS

For iv)(host 6-host10+RTB)

HOST 6 ->199.3.3.5/24

HOST 6 in binary:11000111.00000011.00000011.00000101

Bitwise AND

IP: 11000111.00000011.00000011.00000101

MASK: 11111111.11111111.11111111.11110000

NET:11000111.00000011.00000011.00000000 = 199.3.3.5/24 NETWORK ADDRESS

COLLISION DOMAINS:

Top-left hub (HOST1, HOST2, HOST3, RTA)

All connected to the same hub

Everyone shares the medium

1 collision domain

Top-right hub (HOST4, HOST5, RTA)

Hosts 4 and 5 → connected to a switch

Switch → connected to RTA

Host 4 ↔ switch port → 1 collision domain

Host 5 ↔ switch port → 1 collision domain

Switch ↔ RTA → 1 collision domain

3 collision domains

Bottom left section (HOST6, HOST7, HOST9)

HOST6, HOST7, HOST9-> connected to a hub

Hub-> connected to a switch

HOST6,HOST7,HOST9->HUB->SWITCH

1 collision domain

Bottom right section (HOST 10,RTB)

HOST10<->SWITCH->1 collision

SWITCH<->RTB->1 collision

2 collision domains

Router-to-router link

RTA ↔ RTB

Point-to-point Ethernet link

1 collision domain

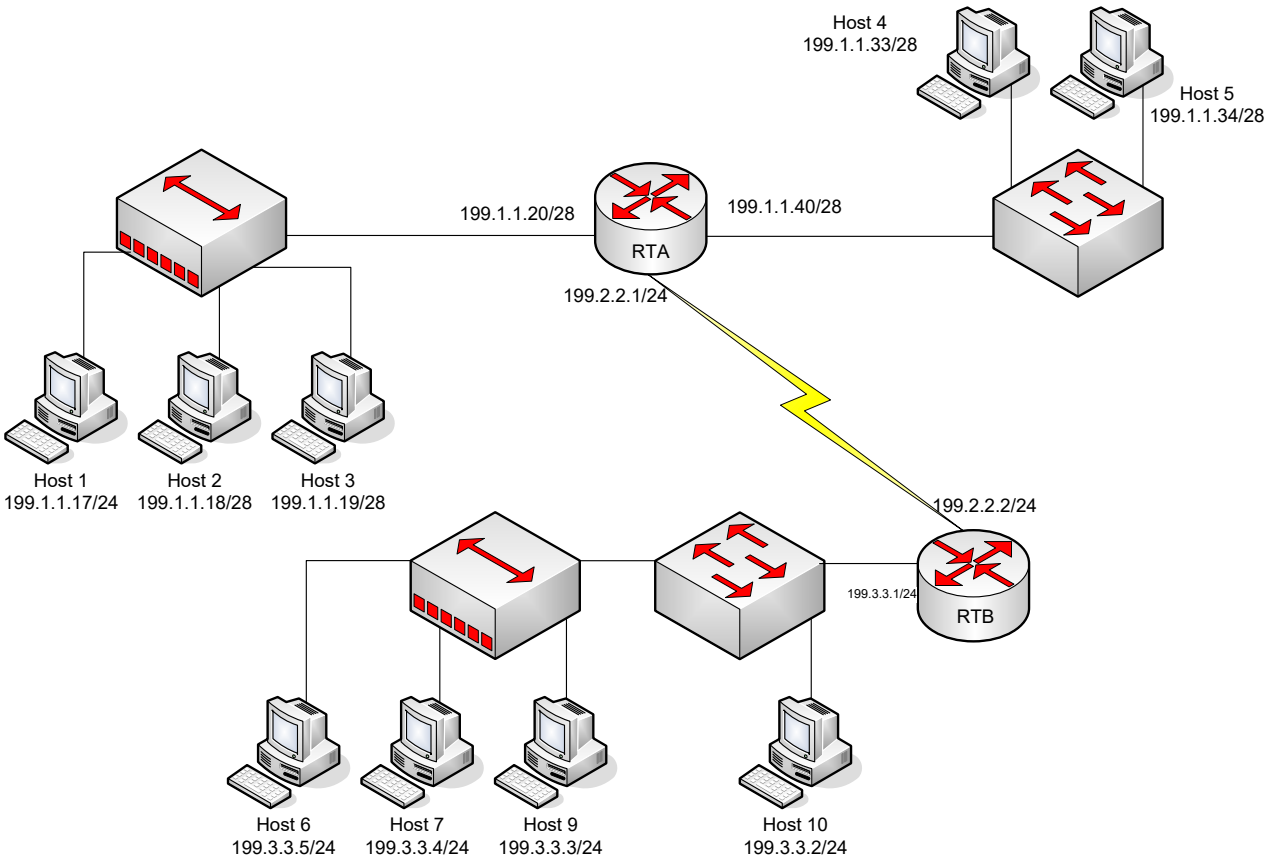
In conclusion there are 8 Collision Domains and 4 Broadcast Domains

7. If we assume that HOST6 wants to communicate with HOST10, for whom he knows the IP address but does not know the corresponding MAC address, describe what exactly must be done and how this will be accomplished, in order finally for these two to be able to communicate. (10 points)

   First HOST6 will check its subnet and see that HOST10 is local. Then HOST6 will send an a broadcast arp request. The switch will flood the ARP request. After that HOST10 will reply with its MAC address. The HOST6 will then store it in the ARP table. Communication will proceed using unicast frames. In conclusion ARP is used to resolve the MAC address, after which normal unicast communication occurs.

8. If we assume the HOST1 wants to communicate with HOST5, for who he knows the IP address but does not know the corresponding MAC address, describe what exactly must be done and how this will be accomplished, in order finally for these two to be able to communicate. (5 points)

   This time because there are different subnets the gateway will be required.HOST1 sees HOST5 is remote.HOST1 ARPs for its default gateway's MAC .The RTA routes the packet toward HOST5's subnet. RTA ARPs for HOST5's MAC.HOST5 replies. End-to-end communication begins.


Host 4
199.1.1.33/28
Host 5
199.1.1.34/28
199.1.1.20/28
RTA
199.1.1.40/28
199.2.2.1/24
199.2.2.2/24
199.3.3.1/24
RTB
Host 1
199.1.1.17/24
Host 2
199.1.1.18/28
Host 3
199.1.1.19/28
Host 6
199.3.3.5/24
Host 7
199.3.3.4/24
Host 9
199.3.3.3/24
Host 10
199.3.3.2/24

Assignment Instructions

Scenario:

You are working as a network consultant for a company that manages a large educational institution offering computer science training. The institution's infrastructure consists of:

- 500 computers distributed across multiple labs used for training purposes.
- 150 employees who need reliable access to the network for administrative and operational tasks.
- A Blade Server array that hosts critical services and applications.
- A Firewall responsible for Inter-VLAN routing and security.
- Two core switches providing redundancy, to which all floors in the building are connected.

The company requires a network design that ensures high availability, security, and efficiency. Your task is to provide a comprehensive solution addressing the following aspects:

Task:

1. VLAN Design:
    - Propose an appropriate number of VLANs and justify your choices.
    - Define how you would segment the network (e.g., by department, function, or security needs) to optimize both performance and security.
2. IP Addressing Scheme:
    - Recommend an IP addressing plan for all devices (students' lab computers, employees' workstations, servers, etc.).
    - Make sure the addressing is scalable, easy to manage, and supports future growth.
    - Use private IP ranges (RFC 1918) and explain how you'll assign IPs to the VLANs.
3. Network Topology:
    - Design a network topology diagram that illustrates the logical structure of the network.
    - Ensure that your topology incorporates sufficient redundancy to limit single points of failure. Consider the connection between core switches, servers, labs, and employee workstations.
    - Describe how the topology will allow for failover in case of hardware failure (e.g., switch or server failures).
4. Redundancy Considerations:
    - Propose a redundancy strategy to avoid single points of failure, especially at the core switch and server levels.
    - Specify how your design will maintain network availability during hardware or link failures, ensuring minimal disruption for users.

Requirements:

- Written Report: explaining your proposed network design, with detailed justifications for each decision.

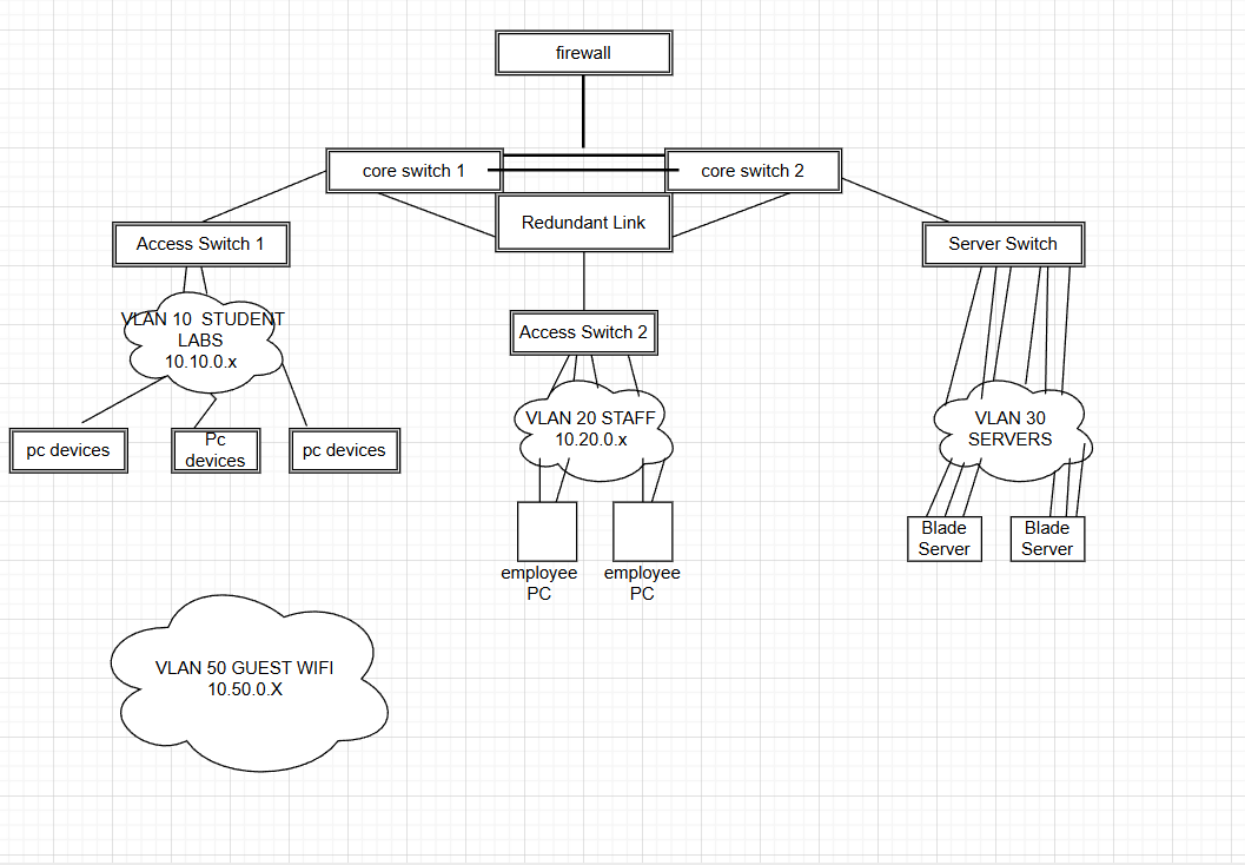
- Network Diagram: A clear, labeled diagram showing your topology, including VLANs, switches, servers, and any other critical devices.
- References: Any resources you consulted should be properly cited.

For this network, I think it's important to separate the different types of users and devices into VLANs. This will help with both performance and security. VLAN 10 Student Labs. This VLAN will include all the computers used for training around 500 computers. Keeping them separate from staff computers prevents the network from slowing down because of the big amount of students. VLAN 20 Staff. All employee workstations go here. It keeps sensitive staff information separate from students and it will be around 150 devices. VLAN 30 Servers. All servers like the Blade Server array and backups, will be on this VLAN. Isolating servers makes them safer and easier to manage and it will be around 20 to 30 devices. VLAN 40 Management. This VLAN is for network devices like switches, printers, and wireless access points. Keeping management devices separate protects the network and it will be around 20 devices. VLAN 50 Guest Wi-Fi. Any visitors can use this VLAN. They will have internet access but won't see the internal network and there wont be a fixed number of devices. Separating devices into VLANs helps the network run faster, makes it more secure, and makes it easier to add more devices in the future.

There will be used private IP addresses like the ones in RFC 1918 for all devices. Here's a plan that is easy to manage and can grow if needed:

Student_Labs (VLAN 10): 10.10.0.0/22 enough addresses for all lab computers.
Staff(VLAN 20): 10.20.0.0/24 enough for 150 employees with room for new staff.
Servers (VLAN 30): 10.30.0.0/25 only for servers to keep them separate.
Management (VLAN 40): 10.40.0.0/26 for switches, printers, and access points.
Guest Wi-Fi (VLAN 50): 10.50.0.0/23 for visitors' devices.

Also, students and guests get their IP addresses automatically using DHCP. Staff computers, servers, and network devices get static IPs, so they don't change. This way the system is simple and can easily handle more devices in the future.



To make sure the network is reliable and always available there should be 2 core switches. Two switches connected together. If one fails, the other takes over. Multiple connections to switches and backup servers for critical services. Access switches each switch has two uplinks to the cores. If one link fails, traffic goes through the other. Firewall high availability pair so one firewall can take over if the other stops working.

This network design keeps students, staff, servers, and guests separate using VLANs. It uses private IP addresses that are easy to manage and can grow. It includes redundancy at critical points to avoid downtime. It's secure, efficient, and scalable for future growth.

5. Deliverable

For this assessment you must deliver your answer as a PDF file. The PDF file should contain
i. Cover page with your name, surname, your University of Greater Manchester ID, Course title and Assessment's title.
ii. Contents
iii. Main part, where you should have your answers as text and not as screenshot images.
iv. References (Harvard style), where you will include all sources which you used, e.g. Your professor's online material, online books, websites, etc.

6. Criteria for grading

For the first part the criteria are alongside each one of the questions.
For the second part:

- VLAN Design (25%): How well does the VLAN structure optimize performance, security, and manageability?
- IP Addressing (25%): Is the IP addressing scheme logical, scalable, and does it account for future growth?
- Network Topology (30%): Does the proposed topology provide sufficient redundancy and ensure high availability?
- Redundancy Strategy (20%): Are redundancy measures well thought out and do they effectively prevent downtime?

7. Learning Outcomes

LO1: Justify an infrastructural strategy to support service requirements based on a given brief
LO2: Apply networking principles and protocols to solve a given problem

8. Use of Generative Artificial Intelligence (GAI) Applications in this Assessment

AI Status	Application	Notes
Category A	No GAI tool is permitted.	While grammar and/or spell checkers may be used to correct individual words and sentences, the use of GAI is not allowed. This is because the learning outcomes require you to produce original assessment work without any GAI assistance Any GAI generated content which is presented as your own original work and is not acknowledged will be assessed for academic misconduct.

9. General Assessment Criteria for Written Assessments

GENERAL ASSESSMENT GUIDELINES – LEVEL HE4

	Relevance **Learning outcomes must be met for an overall pass**	Knowledge and Understanding	Analysis, Creativity and Problem-Solving	Self-awareness and Reflection	Research/ Referencing	Written English	Presentation and Structure
Class I (Exceptional Quality) **85% - 100%**	Work is directly relevant and expertly addresses the requirements of the brief. **Learning outcomes are met.**	Demonstrates an exceptional breadth and depth of knowledge and understanding of theory and practice beyond the threshold expectation for the level. Demonstrates sophisticated conceptual understanding of specialised areas.	Presents a perceptive synthesis and critical evaluation of findings' drawing on a broad range of relevant sources in order to draw clear, systematic, justified conclusions. Provides a critical insight and interprets complex matters and ideas comprehensively. Demonstrates creative flair and a high level of originality. Demonstrates exceptional problem- solving skills and initiative.	Provides insightful reflection and critical self-awareness in relation to the outcomes of own work and personal responsibility.	A wide range of contemporary and relevant reference sources selected and drawn upon. Sources cited accurately in both the body of text and in the Reference List/ Bibliography.	Writing style is clear and appropriate to the requirements of the assessment. An exceptionally well written answer with competent spelling, grammar and punctuation. For example, paragraphs are well structured and include linking and signposting. Sentences are complete and different types are used. A wide range of appropriate vocabulary is used.	The presentational style and layout are correct for the type of assignment. Evidence of planning and logically structured. Where relevant, effective inclusion of, and reference to, figures, tables and images.
Class I (Excellent Quality) **70% - 84%**	Work is relevant and comprehensively addresses the requirements of the brief. **Learning outcomes are met.**	Demonstrates an excellent breadth and depth of knowledge and understanding of theory and practice for this level. Demonstrates in-depth understanding of key areas in different contexts.	Presents an excellent appraisal of findings through the critical analysis of information sources. Draws clear, systematic, justified and perceptive conclusions. Demonstrates creative flair, a high level of originality and initiative. Demonstrates a critical understanding of problem-solving approaches and applies excellent problem-solving skills.	Provides excellent reflection and self-awareness in relation to the outcomes of own work and personal responsibility.	A wide range of relevant reference sources selected and drawn upon. Sources cited accurately in both the body of text and in the Reference List/ Bibliography.	Writing style is clear and appropriate to the requirements of the assessment. An excellently well written answer with competent, spelling, grammar and punctuation. For example, paragraphs are well structured and include linking and signposting. Sentences are complete and different types are used. A wide range of appropriate vocabulary is used.	The presentational style and layout are correct for the type of assignment. Evidence of planning and logically structured Where relevant, effective inclusion of, and reference to, figures, tables and images.
Class II/i (Very Good Quality) **60% - 69%**	Work is relevant and addresses most of the requirements of the brief. **Learning outcomes are met.**	Demonstrates a thorough breadth and depth of knowledge and understanding of theory and practice for this level. Demonstrates very good understanding of key areas in different contexts.	Presents a perceptive appraisal of findings through the critical analysis of information. Draws clear, justified and thoughtful conclusions. Demonstrates a very good level of creativity and originality. Demonstrates a critical understanding of problem-solving approaches and applies strong problem-solving skills.	Provides very good reflection and self-awareness in relation to the outcomes of own work and personal responsibility, as required by the assessment.	A range of appropriate reference sources selected and drawn upon. Sources cited accurately in the main both the body of text and in the Reference List/Bibliography.	Writing style is clear and appropriate to the requirements of the assessment. A very well written answer with competent spelling, grammar and punctuation. For example, paragraphs are well structured and include linking and signposting. Sentences are complete and different types are used. A range of appropriate vocabulary is used.	The presentational style and layout are correct for the type of assignment. Evidence of planning and logically structured in the main. Where relevant, effective inclusion of, and reference to, figures, tables and images.
	Relevance **Learning outcomes must be met for an overall pass**	Knowledge and Understanding	Analysis, Creativity and Problem-Solving	Self-awareness and Reflection	Research/ Referencing	Written English	Presentation and Structure

Class II/ii (Good Quality) **50% - 59%**	Work addresses key requirements of the brief. Some irrelevant content. **Learning outcomes are met.**	Demonstrates a sound breadth and depth of knowledge and understanding of theory and practice for this level. Demonstrates sound understanding of key areas in different contexts.	Presents a logical appraisal of findings through the critical analysis of information. Draws clear and justified conclusions. Demonstrates creativity and originality. Demonstrates a critical understanding of problem-solving approaches and applies effective problem-solving skills.	Provides valid reflection and self-awareness in relation to the outcomes of own work and personal responsibility, as required by the assessment.	Relevant reference sources selected and drawn upon. Most sources accurately cited both the body of text and in the Reference List/Bibliography.	Writing style is mostly appropriate to the requirements of the assessment Grammar, spelling and punctuation are generally competent and minor lapses do not pose difficulty for the reader. Paragraphs are structured and include some linking and signposting. Sentences are complete. A range of appropriate vocabulary is used.	The presentational style and layout are correct for the type of assignment. Logically structured in the most part. Inclusion of figures, tables and images but not all relevant or referred to.
Class III (Satisfactory Quality) **40% - 49%**	Work addresses the requirements of the brief, although superficially in places. Some irrelevant content. **Learning outcomes are met.**	Demonstrates a sufficient breadth and depth of knowledge and understanding of theory and practice for this level. Demonstrates understanding of a range of key areas in different contexts.	Presents valid appraisal of findings in parts following a critical appraisal of information. Interprets a number of matters and ideas but with some descriptive passages evident which lack clear purpose. Demonstrates creativity and originality in places. Demonstrates a critical understanding of problem-solving approaches and applies sufficient problem- solving skills.	Provides some reflection and self-awareness in relation to the outcomes of own work and personal responsibility, as required by the assessment.	Some relevant reference sources selected and drawn upon. Some weaknesses in referencing technique.	Writing style is occasionally not appropriate for the assessment. Grammar, spelling and punctuation are generally competent, but may pose minor difficulties for the reader. Some paragraphs may lack structure, and there is limited linking and signposting. Some appropriate vocabulary is used	The presentational style and layout are largely correct for the type of assignment. Adequately structured. Inclusion of some figures, tables and images but not all clear, relevant and/or referred to.
Borderline Fail **35% - 39%**	Work addresses some of the requirements of the brief. Irrelevant and superficial content. **One or more learning outcomes have not been met.**	Demonstrates a lack of knowledge and understanding of theory and practice for this level. Demonstrates insufficient understanding of a range of areas in different contexts.	Presents a descriptive overview of findings following some analysis of information. Some passages lack clear purpose. Demonstrates scant creativity or originality. Demonstrates a lack of critical understanding of problem-solving approaches and insufficient problem-solving skills.	Provides limited reflection and self-awareness in relation to the outcomes of own work and personal responsibility, when required.	Sources selected are limited and/or lack relevance. Poor referencing technique employed.	Writing style is unclear and does not match the requirements of the assessment in question. Deficiencies in spelling, grammar and punctuation makes reading difficult and arguments unclear in places. Paragraphs are poorly structured.	For the type of assignment the presentational style, layout and/or structure are lacking. Inclusion of figures, tables and images but not clear, relevant and/or referred to.
Fail **<34%**	Work does not address the requirements of the brief. Irrelevant and superficial content. **One or more learning outcomes have not been met.**	Demonstrates inadequate knowledge and understanding of theory and practice for this level. Demonstrates insufficient understanding of relevant areas.	Presents a descriptive overview of findings with little consideration of the quality of information drawn upon. Demonstrates little or no creativity. Demonstrates a lack of understanding of problem-solving approaches and limited problem- solving skills.	Provides inadequate reflection and self-awareness in relation to the outcomes of own work and personal responsibility, when required.	An absence of academic and relevant sources selected and drawn upon. Poor referencing technique employed.	Writing style is unclear and does not match the requirements of the assessment in question. Deficiencies in spelling, grammar and punctuation makes reading difficult and arguments unclear. Unstructured paragraphs.	For the type of assignment the presentational style, layout and/or structure are lacking. Inclusion of figures, tables and images but not clear, relevant and/or referred to.